ELSEVIER Journal of Banking & Finance 25 (2001) 1829–1855

Journal of BANKING & FINANCE

www.elsevier.com/locate/econbase

Index arbitrage with heterogeneous investors: A smooth transition error correction analysis

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Received 8 May 2000; accepted 4 September 2000

Abstract

The traditional index arbitrage model assumes a constant threshold mispricing between the futures and cash prices for all investors. Allowing for heterogeneity in investors' transaction costs, objectives, and capital constraints, we model the intraday mispricing of DJIA futures as a smooth transition autoregressive (STAR) process with the speed of adjustment toward equilibrium varying directly with the mispricing. We show that the observed mean reversion in mispricing changes is induced by heterogeneous arbitrageurs, instead of a statistical illusion-infrequent trading of index portfolio stocks. We further use a STAR error correction model to describe the nonlinear dynamics between the DJIA futures and index. This model describes not only which market is more informationally efficient than the other, but also the legging process – the nonsimultaneous establishing of cash and futures positions. © 2001 Elsevier Science B.V. All rights reserved.

JEL classification: G20; G14; G29

Keywords: Index arbitrage; Heterogeneous investors; STAR model

1. Introduction

Index arbitrage is an important issue for US equity markets, yet reasons for the observed index arbitrage opportunities or significant mispricing between

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PII: S0378-4266(00)00162-X

the cash and futures prices are strongly debated. Harris (1989), Kleidon (1992), and Miller et al. (1994) (MMW, hereafter) argue that infrequent trading of S&P 500 index portfolio stocks can induce the illusion of apparent arbitrage opportunities. However, as shown by Sofianos (1993) and Neal (1996), arbitrageurs trade when the mispricing error is sufficiently wide and these trades can be profitable, suggesting that arbitrage opportunities do occur frequently. Deviations of prices from their arbitraged values arise because information does not flow at the same rate toward both markets. Kawaller et al. (1987), Stoll and Whaley (1990), and Chan (1992), among many others, have reported that the S&P futures price Granger-causes the index price, indicating that the futures market disseminates information more rapidly than the stock market.

The basic index arbitrage models assume a constant mispricing threshold between the futures and cash prices. All investors have the same trading costs and market constraints and they exploit the arbitrage opportunities simultaneously whenever the mispricing exceeds the threshold. In contrast, this paper considers that the trading costs and objectives, perceived risks of arbitrage, and capital constraints vary among investors. What may be a perfectly arbitraged futures price for one investor will be mispriced from the point of view of another. More importantly, "at every futures price there is at least one market participant who should be using this market" (Kawaller, 1991, p. 460).

In this paper, we examine the intraday index arbitrage of the Dow Jones industrial average (DJIA) for the period June 1998–March 1999. The DJIA is better than the S&P 500 for studying the arbitrage mechanism for several reasons. First, since all the thirty Dow stocks are actively traded on the New York Stock Exchange (NYSE), the infrequent trading problem will be mitigated. [1] Second, the size of the DJIA futures contract traded on the Chicago Board of Trade (CBOT) is approximately one-third the size of S&P 500 futures. These specifications scale DJIA stock index arbitrage programs down to a more accessible size and reduce tracking errors and transaction costs. The smallest efficient arbitrage program consists of three DJIA futures and 120 shares of DJIA stock with a value of approximately $260,000. Third, arbitrage orders for the thirty Dow stocks can be submitted simultaneously to SuperDot of the NYSE and will be transmitted to individual stock specialists for execution, [2] resulting in less execution risk. In contrast, S&P 500 arbitrageurs often use surrogate stock baskets containing only 280 stocks, on average, in the cash leg (Sofianos, 1993).

[1] Microsoft and Intel, added to the DJIA in November 1999, are traded on the Nasdaq.

[2] An alternative automated routing and execution to the NYSE is the Chicago Stock Exchange's execution system, MAX (see Shalen, 1999).

We model the intraday mispricing of CBOT DJIA futures as a smooth transition autoregressive (STAR) process of Teräsvirta (1994) and Granger and Teräsvirta (1993). The speed of adjustment toward equilibrium varies directly with the mispricing. More specifically, heterogeneity among investors suggests that the proportion of investors who would capitalize on the mispricing gradually increases with the extent of mispricing. This results in a smooth transition between regimes with strong aggregate adjustment and those with weak adjustment.

The STAR model can explain the puzzle addressed by MMW that significant negative autocorrelation in basis and mispricing changes still exists even after eliminating all pairs of consecutive price changes in which the value of mispricing error exceeds the threshold of the transaction cost bands. According to our simulation experiment, if mispricing follows a smooth transition process, we shall still observe significant negative autocorrelation in mispricing changes even when we eliminate the observations outside the bands. Thus, we can maintain the argument that mean reversion in mispricing is induced by the trading activities of heterogeneous arbitrageurs, instead of a statistical illusion-infrequent trading.

We further use a STAR error correction model to describe the nonlinear dynamics between the DJIA futures and index. Using the linear error correction (EC) model of Engle and Granger (1987), several papers, e.g., Ghosh (1993), Wahab and Lashgari (1993), and Brenner and Kroner (1995), consider the error correction term as the mispricing error. If the futures market reacts to news first, the index will adjust to the disequilibrium more than the futures. Therefore, the error correction term should be more significant in the index than the futures. Incorporating *homogenous* transaction costs among all investors into the EC model has received substantial attention recently. Yadav et al. (1994), Dwyer et al. (1996), Martens et al. (1998), and Forbes et al. (1999) use these threshold autoregressive (TAR) EC models to examine the nonlinear dynamics between the S&P index and futures.

One problem in applying TAR models is the key assumption about the discrete nature of the process. The adjustment mechanism is either entirely in one regime or the other with sharp transition at a common threshold. The fact that different investors could have different transaction cost thresholds suggests that these thresholds will mingle as we aggregate over different investors. Even if they have similar transaction costs, the investors may not react simultaneously to a given mispricing because of different trading objectives, capital constraints, and perceived risks of arbitrage. Therefore, a smooth transition model should be more appropriate to capture the aggregate behavior. This argument parallels the consumption adjustment process described in Marshall and Parekh (1999). They introduce consumption adjustment costs into a simple consumption-based capital asset pricing model (CCAPM) with heterogeneous

agents. They show that individual consumption changes by discrete jumps, while aggregate consumption evolves smoothly. The costs of adjusting consumption among heterogeneous agents can considerably account for the discrepancy between the CCAPM and the observed behavior of asset markets, such as the well known equity premium puzzle reported in Mehra and Prescott (1985).

Despite the rich interpretation of the STAR model, its application in finance has so far been limited. The STAR process is governed by a continuous transition function consisting of two components: a condition that determines when the transition function is active and a parameter that determines the speed of the transition between regimes. The second component, which is not modeled in the linear EC model, plays an important role in describing the index arbitrage mechanism. The STAR EC model describes not only which market reflects information more quickly, but also the legging process – the nonsimultaneous establishing (or liquidating) of cash and futures positions in responding to mispricings. The linear EC model does not distinguish these two processes and may provide biased results. Based on the linear model, we cannot conclude whether the futures or cash market is more efficient in reflecting information. However, the STAR EC model shows that information is reflected in the futures market first and more investors establish positions on the futures market before the stock market. Allowing for asymmetric influence of mispricing, we also find that both markets respond more rapidly to negative mispricing.

The rest of the paper is organized as follows. Section 2 reviews the theory of index arbitrage and discusses the heterogeneity among investors. Section 3 describes the data and frequency of sampling. In Section 4, we analyze the results of mean reversion in mispricing and basis changes. Section 5 provides the theoretical background of the STAR model and examines the empirical results. Section 6 concludes the paper.

2. Heterogeneous arbitrageurs

What the market calls the "fair value" or "break-even price" F_t^* of a futures contract at time t is a futures price that rules out arbitrage between the cash, S_t, and futures, F_t, prices. The fair value is described by the well-known cost-of-carry model:

$$F_t^* = S_t e^{r_t(T-t)} - \sum_{i=1}^{n} D_i e^{r_t(T-\tau_i)}, \tag{1}$$

where T is the expiration day of the futures contract and r_t is the riskless interest rate. Cash dividends represented by D_i with ex-dividend dates prior to T

are invested or discounted at r_t from the payment date τ_i to T. We define the mispricing error as

$$z_t = f_t - f_t^*, \tag{2}$$

where $f_t^* = \ln(F_t^*)$ and $f_t = \ln(F_t)$; thus, we may consider z_t the percentage change of mispricing. We also define the basis as

$$b_t = f_t - s_t, \tag{3}$$

where $s_t = \ln(S_t)$.

Arbitrage involves transaction costs: brokerage costs, shorting costs, bid–ask spreads, and market impact costs when arbitrage trades need to be executed promptly.[3] According to the traditional arbitrage model with homogeneous investors, establishing positions should be a step (or an on/off sharp switch) function of the mispricing; on when $|z_t| > c$, the constant transaction cost threshold, and off otherwise. However, using actual arbitrage trading data on the NYSE, Neal (1996) finds that arbitrage trading is not a step function of mispricing. Rather, larger mispricings are positively related to the frequency of establishing arbitrage trades. These results suggest that different arbitrageurs with different transaction costs face different fair prices and quantities at which they can execute the legs of an arbitrage program. Member firms typically pay no commission fees for proprietary index arbitrage transactions. The fees paid by institutional investors depend on the arrangement between the investor and the executing firm. There does not exist a single fair value beyond which arbitrage becomes profitable for all investors.

For capital-constrained arbitrageurs, the cost of an arbitrage trade includes the opportunity cost of subsequent profitable mispricings that they will be unable to exploit because of their limited capital (Sofianos, 1993). They may skip small mispricings to wait for larger ones. Or they may trade as soon as mispricings appear and forgo larger mispricings when they lack capital. Other than transaction costs and capital constraints, some market frictions and mechanisms described in the following can also explain why investors may not have the same mispricing threshold.

As Kawaller (1987) points out, arbitrageurs have to contend with various risks; see also Sofianos (1993) and Neal (1996). Executions of the cash and futures legs of an arbitrage may be lagged because of delays in the execution of trades. If interest rates are random, index arbitrage is only a quasi-arbitrage, i.e., it is not riskless. The perceived risks of arbitrage vary across investors.

The basic arbitrage model predicts that arbitrage trades will be liquidated at the expiration of the futures contract. However, Sofianos (1993) and Neal

[3] Since institutions are typically net long in the Dow stocks, they can avoid short-sale restrictions by selling the stock directly.

(1996) report that most arbitrage trades are liquidated before the expiration, a finding consistent with the early liquidation option model of Brennan and Schwartz (1990). The model predicts that arbitrage trades will be liquidated when the reversal in the mispricing is sufficiently large. Thus, an arbitrage position will be established when the mispricing plus the value of the option to liquidate the trade early exceeds the transaction costs. The value of this option, which is ignored in the basic model, can be different among investors.

Shleifer and Vishny (1997) further show that arbitrage, in general, is conducted by relatively few specialized professionals. The capital that they invest comes from wealthy outside investors who do not know much about arbitrage and simply use arbitrageurs' performance to ascertain their ability to invest profitably. In their agency model, they show that such specialized performance-based arbitrage may not be fully effective in bringing prices to fundamental values. Accordingly, different agency relationships between arbitrageurs and outside investors will yield different fair values.

The fair value range for a given institution also depends on its motivation, marginal funding, and investing yield alternatives. Kawaller (1991) discusses several differentiated objectives in the futures markets: to generate profits from arbitrage activities, to create synthetic money market instruments, and to change or maintain exposure to equities using the more cost effective instrument via stock/futures substitution. He concludes that no single fair value is universally appropriate. Therefore, assuming a constant mispricing threshold for arbitrage trading among all investors is not practical. In the current paper, we model the arbitrage activities in a smooth transition function of mispricing. Different investors exploit the arbitrage opportunities at different mispricing thresholds, and the aggregate of all the thresholds will become a smooth transition function.

Smooth transition is better than a threshold discrete/sharp switch at describing the aggregate consumption behavior as discussed in Marshall and Parekh (1999). They find that the CCAPM of Lucas (1978), Breeden (1979), and Grossman and Shiller (1982) is sensitive to costs of consumption adjustment, and in their model, different agents have different fixed costs of adjustment.[4] There is a relatively small utility gain from continuous adjustment of consumption in response to movements to asset returns. Agents may forgo this small utility gain because of a minor cost of adjustment. As a result, with adjustment costs, agents only change consumption infrequently and individual consumption is characterized by infrequent discrete jumps. In contrast, the aggregate consumption process evolves smoothly. There does not exist a representative agent (or an overall threshold cost) that can reasonably describe the

[4] More specifically, they assume that each agent's rate of consumption involves a fixed cost, which is proportional to his/her previous rate of consumption.

equilibrium aggregate consumption behavior. Marshall and Parekh show that costs of adjustment consumption and the smooth aggregate process can significantly account for the empirical failure of the CCAPM. In particular, their results explain the discrepancy between the standard deviation of observed nondurable consumption growth and of consumption growth as implied by the CCAPM. They also explains the discrepancy between the actual equity premium observed in the asset markets and the theoretical premium computed from aggregate consumption data.

Another theoretical support for the STAR model can be obtained from Dumas (1992), who examines the dynamic process of the real exchange rate in spatially separated markets under proportional transaction costs. As pointed out by Michael et al. (1997), the adjustment process derived in Dumas (1992) is smooth rather than discrete. Based on Dumas' insights, Michael et al. use the exponential STAR model to describe the adjustment process toward purchasing power parity (PPP), and find strong evidence of mean-reverting behavior for PPP deviations. The STAR model has also been applied in characterizing nonlinearities in business cycles (Teräsvirta and Anderson, 1993), inflation rate (Deutsch et al., 1994), and real effective exchange rates (Sarantis, 1998). Nevertheless, the model has not attracted substantial attention in finance.

3. Data and frequency of sampling

The DJIA is the most widely followed and recognized stock index in the world. The DJIA tracks the average price of 30 large capitalization (blue-chip) stocks – all trade heavily on the NYSE. While the DJIA is the easiest stock index to track, it has not been easy to trade until the advent of the futures contracts traded on the CBOT in October 1997.

The CBOT provides the DJIA time and sales price data for the period June 1998 to March 1999. These data contain the time stamped to the second and price of every futures transaction in which the price has changed from the previously recorded transaction. During this period, we collect three near term futures contracts as follows: Contract September 1998 (Sep 98 in short, hereafter) from 11 June 1998 to 9 September 1998, Contract December 1998 (Dec 98) from 10 September 1998 to 9 December 1998, and Contract March 1999 (Mar 99) from 10 December 1998 to 10 March 1999. As in MMW, the last day of each contract used is at least five trading days before its last trading day. [5]

[5] The minimum price fluctuation (or tick size) in the futures contract is one point, or \$10. Settlement occurs on the third Friday of the contract months, March, June, September, and December. The final settlement price is \$10 times the Special Opening Quotation of the index.

Arbitrage trades are concentrated in the near term contract until a few days before expiration (see Neal, 1996, p. 546). [6]

We calculate the index price, S_t, by summing the NYSE transaction prices of the 30 stocks and dividing that sum by a divisor. The transaction prices are obtained from the NYSE Tick and Quote (TAQ) database. The divisor changes to account for stock splits and stock dividends, and different values of the divisor during the period are provided by Dow Jones & Company. To calculate the fair value of the futures contract, F_t^*, in (1), we need the riskless rate, the individual future cash dividends for the Dow stocks, and the ex-dividend and dividend payment dates. Like Chung (1991), our proxy for the riskless rate is the average of the bid and ask discounts on the Treasury bill (reported in *The Wall Street Journal*) which matures on the day that is closest to the last trading day of the futures contract. Neal (1996) shows that the opportunity cost of arbitrage funds exceeds the Treasury bill rate by 0.88%. Overall results of the current paper are qualitatively the same when we add this percentage to the Treasury bill rate. The dividend information is obtained from Standard & Poor's Stock Reports. Our proxy for the future cash dividends is the actual cash dividends realized over futures' life. Since future dividends of the Dow stocks are fairly predictable, discrepancies caused by this proxy should be small.

The floor trading hours of the CBOT futures are 7:20 a.m. to 3:15 p.m. Chicago time, while the NYSE opens at 8:30 a.m. and closes at 3 p.m. Chicago time. We use the contemporaneous trading hours of the CBOT and NYSE – 9:00 a.m. to 3 p.m. The first 30 minutes of the NYSE are excluded to purge the stale opening prices of the index. The observed arbitrage opportunities during these 30 minutes are illusory because the prices of some component stocks in the index may still be the previous day's closing stock prices. [7] While the trade reporting delay for open outcry trading on the CBOT floor is a few seconds, the delay at the NYSE is about 16 seconds (Blume and Goldstein, 1997). In his study of the information content of program and index-arbitrage orders, Hasbrouck (1996) adjusts the NYSE transactions by 15 seconds. This adjustment, however, does not change our results qualitatively.

The first issue in examining intraday index arbitrage and price discovery is the frequency of the data used. Previous studies on the S&P 500 index have used intervals of 15 minutes (e.g., MMW), 5 minutes (Chan, 1992), and 1 minute (Dwyer et al., 1996). How actively the futures contracts and the

[6] The transaction costs and execution risk of index arbitrage are lower for the near term contract than the longer term contracts because the near term contract is more actively traded.

[7] We also exclude an interval of 15 minutes, 2:15 to 2:29 p.m., on 15 October 1998. During this interval, the announcement of the Federal Reserve's quarter-point cut in overnight lending rates sent stocks and futures climbing and incurred an outlier in each market.

Table 1
Number and percentage of intervals with no trading[a]

	Sep 98	Dec 98	Mar 99
Panel A: 1-minute interval			
Sample size of all intervals	22 680	22 570	21 780
Number (Percentage) of intervals with no trading			
Futures	604 (2.66)	1346 (6.96)	1509 (6.93)
IBM	1958 (8.63)	1288 (5.70)	944 (4.33)
Panel B: 5-minute interval			
Sample size of all intervals	4536	4514	4356
Number (Percentage) of intervals with no trading			
Futures	1 (0.00)	4 (0.09)	5 (0.11)
IBM	0 (0.00)	0 (0.00)	0 (0.00)

[a] This table reports the number and percentage of 1-minute and 5-minute intervals during which no trading has occurred. The first row of each panel presents the sample size of all intervals.

underlying stocks trade determines the sampling interval. DJIA futures are actively traded with an average daily trading volume of 14 thousand contracts in the first quarter of 1999, but still much less than the 92 thousand contracts of the S&P 500 futures.

Table 1 reports the number and percentage of 1-minute and 5-minute intervals during which no trading has occurred on the futures market. Like other tables hereafter, results of each contract/period (Sep 98, Dec 98, and Mar 99) are reported. The table also presents the results of IBM, which has the greatest weight in the DJIA. The average sample size of the three contracts for all the 1-minute and 5-minute intervals are 22,343 and 4469, respectively. Panel A shows that, on average, 5.2% of the 1-minute intervals had no futures trading.[8] For this sampling, IBM did not trade during 7.6% of the intervals. This infrequent trading problem should be more severe for other Dow stocks, which are less actively traded than IBM. Using 5-minute sampling, Panel B shows that both the futures contracts and IBM shares were traded almost in every consecutive interval (or 0% nontrading intervals) for the entire period. Thus, it appears that the futures and cash markets are active enough to adopt the 5-minute, but not 1-minute, interval frequency.[9]

[8] These percentages are larger than the true percentages because if the transaction price is the same as the previous one, no transaction price is recorded.

[9] DIAMONDS, traded on the American Stock Exchange (AMEX), offer an alternative arbitrage against CBOT DJIA futures. DIAMONDS shares give investors ownership in the Dow stocks with a single stock trade. However, the shares are not as actively traded as the futures. For example, DIAMONDS shares are not traded on 15.1% of the 5-minute intervals during the period examined.

4. Mean reversion in basis and mispricing changes

Table 2 presents the first eight-orders of autocorrelations for the 5-minute changes of futures, index, basis, and mispricing. Overnight changes are dis-

Table 2
Autocorrelations of changes of futures, index, and basis[a]

	Sep 98	Dec 98	Mar 99
Panel A: Futures, Δf_t			
First-order	−0.058 (−2.00)	−0.137 (−6.32)	−0.128 (−5.61)
2	−0.053 (−1.79)	−0.032 (−1.43)	0.015 (0.86)
3	−0.025 (−0.94)	0.032 (1.44)	0.010 (0.60)
4	0.026 (0.93)	−0.005 (−0.23)	−0.001 (−0.06)
5	0.009 (0.30)	−0.008 (−0.37)	−0.010 (−0.06)
6	0.035 (1.30)	−0.005 (−0.27)	0.002 (0.10)
7	0.034 (1.25)	−0.030 (−1.45)	0.011 (0.63)
8	−0.018 (−0.70)	−0.005 (−0.27)	−0.001 (−0.07)
Panel B: Index, Δs_t			
First-order	0.101 (3.43)	−0.040 (−1.93)	−0.029 (−1.69)
2	−0.041 (−1.33)	−0.008 (−0.35)	−0.006 (−0.35)
3	−0.038 (−1.30)	0.044 (2.10)	0.004 (0.22)
4	0.000 (0.01)	−0.012 (−0.61)	−0.007 (−0.39)
5	0.033 (1.05)	−0.025 (−1.20)	−0.006 (−0.37)
6	0.045 (1.66)	−0.001 (−0.06)	−0.007 (−0.39)
7	0.017 (0.70)	−0.041 (−1.97)	0.008 (0.48)
8	0.010 (0.40)	0.007 (0.35)	−0.003 (−0.20)
Panel C: Basis, Δb_t			
First-order	0.432 (14.55)	−0.452 (−20.17)	−0.457 (−19.60)
2	−0.080 (−2.74)	0.022 (1.16)	0.008 (0.49)
3	0.012 (0.38)	−0.029 (−1.54)	−0.022 (−1.25)
4	0.008 (0.31)	0.006 (0.35)	0.013 (0.76)
5	0.018 (0.70)	0.019 (0.85)	−0.010 (−0.61)
6	0.014 (−0.57)	−0.033 (−1.57)	−0.007 (−0.42)
7	0.018 (0.65)	0.035 (1.76)	0.015 (0.93)
8	−0.012 (−0.47)	−0.030 (−1.68)	−0.027 (−1.63)
Panel D: Mispricing, Δz_t			
First-order	−0.431 (−14.54)	−0.452 (−20.17)	−0.457 (−19.61)
2	−0.080 (−2.75)	0.022 (1.18)	0.008 (0.48)
3	0.011 (0.38)	−0.029 (−1.54)	−0.022 (−1.26)
4	0.008 (0.31)	0.006 (0.35)	0.013 (0.77)
5	0.018 (0.70)	0.019 (0.85)	−0.010 (−0.61)
6	−0.014 (−0.57)	−0.033 (−1.57)	−0.007 (−0.42)
7	0.018 (0.65)	0.035 (1.77)	0.015 (0.93)
8	−0.012 (−0.47)	−0.030 (−1.68)	−0.027 (−1.64)

[a] We define the basis and mispricing as $b_t = f_t - s_t$ and $z_t = f_t - f_t^*$, respectively, where f_t, s_t, and f_t^* are the futures, cash, and fair prices in logarithm. The fair price is described by the well-known cost-of-carry model. t-statistics in the parentheses are derived from the Diebold (1988) ARCH-adjusted convariance matrix.

carded. Panel A shows that the futures price changes exhibit a moderately negative first-order autocorrelation, −0.06, −0.14, −0.13, for the Sep 98, Dec 98, and Mar 99 contracts, respectively. All *t*-statistics reported in the parentheses are calculated by the Diebold (1988) ARCH-adjusted covariance. This negative autocorrelation can be explained by the bid–ask price bounce (Roll, 1984). Panel B reports the autocorrelations of index changes. The results of the Dec 98 and Mar 99 periods are insignificant, −0.04 and −0.03, while the Sep 98 period has a value of 0.10. Such a low value of autocorrelation suggests that infrequent trading of stocks within the index portfolio is insignificant. (If 1-minute intervals are used, the average first order autocorrelation is significantly positive, 0.46, indicating severe infrequent trading in the index for a higher frequency sampling.)

We present the autocorrelations of basis changes and mispricing changes, respectively, in Panels C and D of Table 2. The results of these panels are almost identical: significantly negative first order autocorrelation for all the three periods, −0.43, −0.45, and −0.46. Autocorrelations beyond the first-order are insignificant. Lengthening the interval to 15 minutes does not reduce these negative first order autocorrelations, −0.44, −0.38, and −0.42. These results should not be explained by the infrequent trading in the index, which demonstrates a small autocorrelation at the 5-minute interval.

Numerous papers have reported mean reversion in intraday stock index basis and mispricing changes. MacKinlay and Ramaswamy (1988) find that the average first-order autocorrelation is −0.23 for the period 1983–1987 and MMW report a value of −0.37 for the period 1982–1991. Both papers use the S&P 500 index. This mean reversion is consistent with the effects of arbitrage. If the basis widens substantially (i.e., the futures price is too high relative to the index value), arbitrageurs sell index futures and buy the index portfolio. If the basis narrows significantly, they will do the reverse. These trades drive prices back to equilibrium.

An alternative explanation for the observed mean reversion is infrequent trading of the index. MMW argue that basis changes would be mean-reverting as lagging stocks in the index eventually trade and their prices are updated, even if no arbitrage has occurred. In particular, Proposition 3 of MMW (p. 498) shows that the first-order autocorrelation in observed basis changes is always negative under this infrequent trading problem and several weak assumptions. Significant negative autocorrelation in observed basis and mispricing changes still exists even after eliminating all pairs of consecutive price changes in which the absolute value of the mispricing exceeds the presumed arbitrage transaction costs (0.25%). MMW consider mispricings outside these transaction bands to be potential arbitrage opportunities, and, therefore, conclude that arbitrage cannot explain the mean reversion in basis and mispricing changes.

Figlewski (1984), Brennan and Schwartz (1990), Kumar and Seppi (1994), Fremault (1991), and Holden (1990, 1995) have also modeled the mispricing as

a mean-reverting stochastic process derived from short-lived information differences or stochastic order imbalances due to market-specific trading clienteles and market maker inventory costs. In particular, Kumar and Seppi consider the role of index arbitrage in the transmission of information across markets and its impact on the basis. [10]

We examine the first-order autocorrelation of basis and mispricing changes by following MMW's approach. Autocorrelations in basis changes are almost identical to those of mispricing changes; therefore, we only report the results of basis changes. Table 3 presents the autocorrelations when all pairs of consecutive price changes in which the absolute value of the mispricing ranges from 0.10% to 0.60% (with a step increase of 0.025%) are eliminated. For the 0.20% or above transaction bands, the autocorrelations are similar to the original values. Even at a small transaction band of 0.10%, with more than half of the observations eliminated, the first-order autocorrelation is still significantly negative for each period, −0.27, −0.39, and −0.37. Why does mean reversion in basis and mispricing changes still exist when so many observations have been discarded by such a small filter? If we maintain the notion that arbitrage is the reason for the mean reversion, arbitrage should not occur infrequently at a clear-cut threshold point of mispricing. Instead, arbitrage trades should be frequently transacted with a wide spectrum of mispricing.

MMW also find that the ratio of the variance of futures changes to variance of index changes, VR, is greater than 1. They show that this result is consistent with their Proposition 1 that the infrequent trading of stocks can cause VR to exceed 1. However, that $VR > 1$ is also consistent with the result that the futures market impounds information more rapidly than the cash market. Following Ross' (1989) model, variance is directly related to information flow. In the current study, we find that the variance ratios of the three contracts are all significantly greater than 1, 1.41, 1.46, and 1.30.

4.1. Simulation experiment on mean reversion

In the following, we support the use of the STAR model in index arbitrage by conducting a Monte Carlo experiment. We simulate the mispricing series z_t as the following threshold model:

$$z_t = \begin{cases} z_{t-1} + e_t & \text{if } |z_{t-d}| \leqslant c, \\ \varphi z_{t-1} + e_t & \text{if } |z_{t-d}| > c. \end{cases} \tag{4}$$

The delay parameter d is set to 1. Drawings of the pseudo-random innovations $e_t, t = 1, \ldots, T + 500$, are performed from the standard distribution, $N(0, 1)$.

[10] Hasbrouck (1996) and Laux and Furbush (1993) provide empirical evidence that index-arbitrage orders contain information beyond that available from the futures return and basis.

Table 3
First-order autocorrelations of basis changes: 5-min interval[a]

Mispricing	Sep 98		Dec 98		Mar 99	
	Sample size	Autocorr.	Sample size	Autocorr.	Sample size	Autocorr.
0.100	1063	−0.264	1075	−0.394	1858	−0.370
		(−7.58)		(−10.10)		(−13.70)
0.125	1592	−0.366	1499	−0.413	2347	−0.421
		(−11.49)		(−12.47)		(−16.37)
0.150	2177	−0.365	1927	−0.429	2798	−0.438
		(−13.00)		(−14.34)		(−18.15)
0.175	2789	−0.390	2360	−0.421	3160	−0.457
		(−15.31)		(−15.26)		(−19.43)
0.200	3358	−0.412	2730	−0.439	3472	−0.459
		(−16.57)		(−17.88)		(−20.38)
0.225	3816	−0.435	3096	−0.446	3748	−0.451
		(−18.01)		(−19.08)		(−21.10)
0.250	4147	−0.450	3410	−0.436	4000	−0.457
		(−19.08)		(−19.65)		(−21.45)
0.275	4327	−0.442	3693	−0.441	4139	−0.459
		(−18.96)		(−20.24)		(−22.09)
0.300	4424	−0.455	3916	−0.455	4244	−0.461
		(−18.27)		(−21.02)		(−22.59)
0.325	4484	−0.453	4102	−0.452	4299	−0.459
		(−18.49)		(−21.99)		(−22.69)
0.350	4511	−0.450	4248	−0.454	4329	−0.458
		(−18.23)		(−21.99)		(−22.82)
0.375	4517	−0.449	4367	−0.463	4341	−0.458
		(−18.22)		(−22.64)		(−22.82)
0.400	4521	−0.448	4432	−0.475	4348	−0.458
		(−18.31)		(−21.13)		(−22.82)
0.425	4527	−0.445	4471	−0.475	4351	−0.455
		(−18.10)		(−21.79)		(−22.78)
0.450	4527	−0.445	4488	−0.477	4352	−0.455
		(−18.10)		(−21.71)		(−22.77)
0.475	4527	−0.445	4490	−0.477	4352	−0.455
		(−18.10)		(−21.71)		(−22.77)
0.500	4529	−0.450	4500	−0.476	4352	−0.455
		(−16.66)		(−21.30)		(−22.77)
0.525	4529	−0.450	4500	−0.476	4352	−0.455
		(−16.66)		(−21.30)		(−22.77)
0.550	4529	−0.450	4500	−0.476	4354	−0.455
		(−16.66)		(−21.30)		(−22.78)
0.575	4530	−0.448	4502	−0.476	4354	−0.455
		(−16.62)		(−21.22)		(−22.78)
0.600	4532	−0.436	4504	−0.474	4354	−0.455
		(−15.31)		(−21.19)		(−22.78)

[a] This table presents the first-order autocorrelations when all pairs of consecutive price changes in which absolute value of the mispricing exceeds 0.10–0.60% (with a step increase of 0.025%) are eliminated. *t*-statistics in the parentheses are derived from the Diebold (1988) ARCH-adjusted convariance matrix.

The first 500 observations are discarded, giving a usable sample size of $T = 5000$. We simulate model (4) 10,000 times.[11] The model illustrates that if the absolute value of mispricing, $|z_{t-1}|$, is less than or equal to the transaction cost threshold, c, z_t follows a random walk process. But if $|z_{t-1}|$ is greater than c, z_t follows an autoregressive (AR) process, and hence mispricing changes Δz_t will be mean-reverting. Detailed analysis of the threshold autoregressive (TAR) is provided by Tong (1990) and Tsay (1989).

We set $c = 2.5$ (or 2.5 times of the standard deviation of the innovations, e_t) and the AR parameter $\varphi = 0.1$. The conclusions are the same when we use $c = 2.0$ and 3.0 and $\varphi = 0.3$ and 0.5. From the simulations of (4), the average of the first-order autocorrelation of Δz_t, ρ_1, is significantly negative, -0.227 (with a standard deviation, S.D. $= 0.012$). Now we eliminate all pairs of consecutive price changes in which $|z_{t-1}| > c$. The average sample size is reduced to 3972 (S.D. $= 36.1$) and ρ_1 to -0.112 (S.D. $= 0.015$). Thus, if the arbitrage mechanism does follow a threshold process such as assumed in MMW and many other papers, the magnitude of ρ_1 or mean reversion in basis changes and mispricings should be decreased significantly (from -0.227 to -0.112 in this experiment) when we delete all price changes with mispricing outside the transaction bands.

But what will happen if z_t follows a STAR model because of the heterogeneity of investors as discussed in Section 4? See Teräsvirta (1994) and Granger and Teräsvirta (1993) for detailed analysis of the STAR model. The STAR model aggregates different thresholds (or step functions) among all investors to a smooth function:

$$\begin{aligned} z_t &= z_{t-1} + (-z_{t-1}) \times W(z_{t-d}) + e_t, \\ W(z_{t-d}) &= 1 - \exp\{-\lambda z_{t-d}^2\}. \end{aligned} \tag{5}$$

Model (5) is also called the exponential STAR (ESTAR) model. The STAR model allows for the fact that the larger the deviation from equilibrium, the stronger the tendency to move back to equilibrium. z_{t-d} is the transition variable, $d = 1$ is the delay parameter, and $W(\cdot)$ is the response/transition function.

An alternative to the ESTAR is the logistic STAR (LSTAR) model with a different transition function. As pointed out in Michael et al. (1997), the ESTAR model is better than the LSTAR model in examining the adjustment process of deviation from equilibrium. We may consider the ESTAR model (whereby the transition between regimes is smooth) a generalization of the TAR model (whereby the transition is a sharp switch) in describing the error

[11] All simulations are performed by Digital Visual Fortran 6.0 with the IMSL function generating a random normal variate $N(0, 1)$.

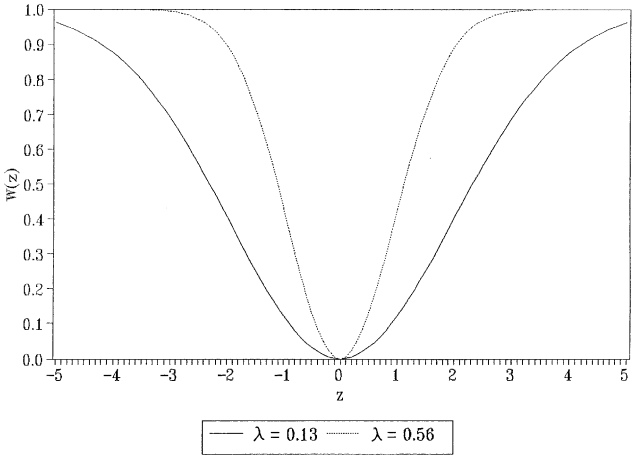


Fig. 1. Transition function.

correction dynamics.[12] The parameter $\lambda > 0$ measures the steepness of the response function or the speed of transition between two regimes: one is close to or at equilibrium (the middle regime) and one is far away from equilibrium (the outer regime). The middle regime corresponds to $z_{t-1} = 0$ when $W = 0$ and z_t follows a random walk process. The outer regime corresponds to $|z_{t-1}| = \infty$ when $W = 1$ and $z_t = e_t$. The transition function varies between zero and one smoothly for different values of z_{t-1}. The greater the λ, the more mean-reverting of Δz_t. We illustrate this in Fig. 1, which plots the transition functions with two different values of λ, 0.13 and 0.56.

To compare the autocorrelation results of the smooth transition model (5) with those of the threshold model (4), we set $\lambda = 0.13$ so that the first-order autocorrelation ρ_1 is the same as that of the threshold model, −0.227. Then we repeat the elimination process with a cut-off point of 2.5. The sample size reduces to 4520 (S.D. = 29.3), but ρ_1 only slightly decreases to −0.210. We also use a smaller cut-off point, 2.05, so that the sample size is the same as the threshold model after trimming, 3972. The value of ρ_1 remains almost unchanged, −0.212. In another simulation setting, we increase to 0.56 so that the average ρ_1 is similar to the empirical value of the mispricing changes reported in Table 2, −0.437. After eliminating all pairs of consecutive price changes in

[12] More specifically, the ESTAR model may be considered a generalization of a TAR model with two thresholds (or three regimes: one within equilibrium, two outside equilibrium), and LSTAR a TAR model with only one threshold (or two regimes: positive and negative deviations from equilibrium).

which $|z_{t-1}| > 2.5$, ρ_1 only decreases minimally to -0.415 (S.D. $= 0.01$) with an average sample size of 4547 (S.D. $= 28.6$).

This simulation experiment shows that if mispricing follows a smooth transition process, we shall still observe significant negative autocorrelation in mispricing changes even when we eliminate a large proportion of observations outside the threshold of the transaction cost band. Thus, we can maintain the argument that mean reversion in mispricing is induced by the trading activities of heterogeneous arbitrageurs.

5. Error correction between markets with heterogeneous arbitrageurs

5.1. Some theoretical background in the linear error correction model

Linear vector error correction (EC) models of Engle and Granger (1987) have been used extensively in describing two or multiple cointegrated economic and financial series. Brenner and Kroner (1995) review various empirical examples in finance. They show how closely linked securities are cointegrated and tied together by arbitrage; see also Hasbrouck (1995).

Consider two cointegrated series x_{1t} and x_{2t}. The usual linear vector EC model will behave as follows:

$$\begin{aligned}\Delta x_{1t} &= a_{10} + \gamma_1 z_{t-1} + \sum_{k=1}^{p} a_{1k}\Delta x_{1,t-k} + \sum_{k=1}^{p} b_{1k}\Delta x_{2,t-k} + e_{1t},\\ \Delta x_{2t} &= a_{20} + \gamma_2 z_{t-1} + \sum_{k=1}^{p} a_{2k}\Delta x_{1,t-k} + \sum_{k=1}^{p} b_{2k}\Delta x_{2,t-k} + e_{2t},\end{aligned} \tag{6}$$

where $z_{t-1} = x_{1,t-1} - x_{2,t-1}$ is the EC term. The parameters of importance are γ_1 and γ_2, which describe the adjustment dynamics between the two markets. If information disseminates in x_{1t} first and x_{2t} later (i.e., x_{1t} leads x_{2t} in the Granger sense), γ_1 should be zero or insignificant because x_{1t} reflects information faster than x_{2t} and, therefore, x_{1t} will not adjust to the disequilibrium described by z_{t-1}. γ_2 is significantly positive because x_{2t} follows x_{1t} to reflect information, and x_{2t} will adjust to the disequilibrium. Similarly, if information disseminates in x_{2t} first, γ_1 should be significantly negative, and γ_2 is insignificant. In sum, the higher the magnitude (with correct sign) of γ_i, the slower the market i reflects information.

The linear EC model does not consider the fact that error correction or adjustment will not occur unless mispricing exceeds the transaction cost threshold. Dwyer et al. (1996) assume a group of arbitrageurs with the same transaction costs and model the adjustment process as a threshold EC model. In the current paper, we assume heterogeneity in investors' trading motivations, capital constraints, and transaction costs. When we aggregate the

thresholds among all investors, the different thresholds will become a smooth transition function. Therefore, at every futures price there is at least one investor who should be using this market.

We use the following exponential STAR EC model to examine the DJIA index arbitrage with 5-minute intervals:

$$\Delta x_{it} = a_{i0} + \sum_{k=1}^{p} a_{ik}\Delta f_{t-k} + \sum_{k=1}^{p} b_{ik}\Delta s_{t-k} + \left(\sum_{k=1}^{p} c_{ik}\Delta f_{t-k} + \sum_{k=1}^{p} d_{ik}\Delta s_{t-k} + \gamma_i z^*_{t-1} \right) \times W_i(z^*_{t-d}) + e_{it}, \tag{7}$$

$$W_i(z^*_{t-d}) = 1 - \exp\{-\lambda_i \times (z^*_{t-d})^2 \times g_i(z^*_{t-d})\}, \tag{8}$$

$$g_i(z^*_{t-d}) = 0.5 + 1/[1 + \exp\{-\theta_i z^*_{t-d}\}], \tag{9}$$

where $i = 1$ specifies the futures market and $i = 2$ the cash market. We obtain the EC term z^*_t by running the following OLS:

$$f_t = c_0 + c_1 f^*_t + c_2 \times DEXP_t + z^*_t, \tag{10}$$

where $DEXP_t$ is the number of days before the expiration of the futures contract. We find that c_1 is almost identical to one and c_2 is significantly positive. (Results of Eq. (10) are not reported for brevity.) That is, z^*_t is the demeaned mispricing between the futures and fair value adjusted for the number of days before expiration. MacKinlay and Ramaswamy (1988) and Klemkosky and Lee (1991) find that mispricing diminishes as expiration of the futures contract approaches.

Suppose a piece of news is announced: e.g., an unexpected decrease in interest rate. The above STAR EC model examines two processes of nonlinear dynamics between the DJIA futures and cash markets. The first process describes which market will react to this information earlier. If the futures price reacts (rises, in this case) first, followed by the cash market, γ_2 should be significant but not γ_1. Since the cash price increases with a lag, mispricing or deviation between the futures and fair prices occurs. If the mispricing is significant, arbitrage is triggered with a short position in futures and a long position in cash. The second process analyzes how rapidly the two markets respond to this misprcing. The proportion of investors who would exploit the mispricing gradually increases with the extent of mispricing. A large (small) λ_i suggests that more (fewer) investors in market i will respond to a given mispricing. Accordingly, the faster the response of market i to mispricing, the higher value of λ_i.

Arbitrage theory assumes that the stock and futures positions or legs are established simultaneously. However, in practice, one position may be submitted a few minutes before the other – legging. Neal (1996) finds that about 81% of S&P 500 arbitrage trades have the two positions submitted

simultaneously. The futures position is submitted before the stock in 16% of the trades, but the stock position is submitted before the futures in only 3.2%. Sofianos (1993) reports similar results: In 22% of trades, the arbitrageur establishes the futures leg more than 5 minutes after the cash leg, but 12% of trades establish the cash leg first. One reason for an arbitrageur to establish the futures position first is that the futures price is more volatile than the index. Submitting the futures position first can provide him/her with time to obtain a more favorable price. An alternative explanation is that the futures position can be liquidated quickly if something goes wrong. These two factors will result in a larger value of λ_1 than λ_2 in the response function (8).

Function $g_i(\cdot)$ in Eq. (9) accounts for the asymmetric response of investors to disequilibrium (Anderson, 1997). If θ_i is negative (positive), then more investors will respond to a negative (positive) mispricing, z^*_{t-1} at time t, than to a positive (negative) mispricing of the same magnitude. If $\theta_i = 0$ and $g_i(\cdot) = 1$, then investors respond to negative and positive mispricings symmetrically. Therefore, $g_i(\cdot)$ can examine whether investors will respond differently when the futures price is below or above the fair value.

Before we estimate the exponential STAR EC model, it is important to test and reject the null hypothesis of linearity, i.e., $H_0 : \lambda_i = 0$ against $H_1 : \lambda_i > 0$, because the model is only identified under the alternative hypothesis. Following the approach of Davies (1977), Teräsvirta (1994) develops a Lagrange multiplier-type test to test linearity against the STAR models. See Michael et al. (1997, pp. 867–868) for a summary of this test for the exponential STAR models. The linearity test is performed for different values of the delay parameter d. If linearity is rejected for more than one d, we choose the one for which the p-value of the test is the lowest.[13] For all futures and cash series, linearity is rejected in favor of the exponential STAR model (with p-value <0.001) and $d = 1$ is chosen. Detailed results of the linearity test are available upon request.

Having rejected linearity, we proceed to estimate the STAR model. Following Teräsvirta (1994) and most previous studies using the STAR models, we standardize λ_i and θ_i as λ_i/σ_i^2 and θ_i/σ_i for better convergence, where σ_i^2 is the unconditional variance of series i. We allow the residuals in Eq. (7) to follow an exponential GARCH (EGARCH) process of Nelson (1991), $e_{it} \sim N(0, \sigma_{it}^2)$:

$$\sigma_{it}^2 = \exp\{\omega_i + \alpha_i(|u_{i,t-1}| - E|u_{i,t-1}| + \delta_i u_{i,t-1}) + \beta_i \ln(\sigma_{i,t-1}^2)\}, \tag{11}$$

where $u_{it} = e_{it}/\sigma_{it}$ is the standardized residual. The asymmetric volatility coefficient δ_i accommodates the observation that volatility tends to rise more with

[13] The justification behind this approach is that the linearity test has the highest power against the alternative when d is selected correctly.

bad news (or negative previous innovations) than with good news (or positive previous innovations).[14] Consistent with this observation, Nelson finds that δ is significantly negative in the US stock market (see also Christie, 1982; Black, 1976). In this paper, we do not consider volatility spillovers (or meteor showers as labeled by Engle et al., 1990) between markets. Several papers, e.g., Chan et al. (1991), Koutmos and Tucker (1996), and Tse (1999), have examined how the volatility of the stock market will affect that of the futures market, and vice versa.

We estimate Eqs. (7)–(9) and (11) simultaneously for each market; i.e., $i = 1$ or 2 in the equations each time by maximizing the log likelihood:

$$l(\Theta) \propto \sum_{t=1}^{T} \left[-\ln(\sigma_{it}^2) - e_{it}^2/\sigma_{it}^2 \right], \tag{12}$$

where Θ is the vector of parameters. For such a nonlinear system, it is impractical to estimate the two markets simultaneously. To the best of our knowledge, no previous work has jointly estimated the STAR models with more than one series. Indeed, even the univariate estimation process is not an easy task. To obtain the initial values for the maximum likelihood estimation, we use a nonlinear least squares procedure with a grid search over the nonlinearity parameters λ_i and θ_i. Different initial values have been tried because the likelihood function can have several local maximums.

We conduct a Monte Carlo experiment to investigate whether results of the nonlinear STAR model will be biased if the model is not jointly estimated. Investigating this issue with a closed-form solution seems infeasible. We simulate a simple STAR EC model with no lagged futures and cash returns (i.e., $p = 0$), symmetric response to positive and negative mispricing (i.e., $\theta = 0$), $\lambda = 0.13$, conditional normality, and three different settings of γ_i ($\gamma_1 = -0.1$ and $\gamma_2 = 0.1$; $\gamma_1 = 0.0$ and $\gamma_2 = 0.1$; and $\gamma_1 = -0.1$ and $\gamma_2 = 0.0$). The first setting assumes that the two series Granger-cause each other equally. The second (third) setting assumes that the first (second) series Granger-causes the second (first) series, but not vice versa. For all the settings, the residuals of the two series in this simple STAR EC model are contemporaneously related with a correlation coefficient of 0.9. The technical issues of simulations (e.g., a usable sample size of 5000) are similar to the first experiment in Section 4. We find that the estimation results (i.e., γ and λ) are similar whether we estimate the two series separately or jointly. To save space, the results are not reported, but are available upon request.

[14] Kim and Kon (1994) compare different models for the conditional heteroscedasticity of stock returns. They find that the EGARCH is the most descriptive for stock indexes.

Table 4
Estimations of STAR EC models [a,b]

$$\Delta x_{it} = a_{i0} + \sum_{k=1}^{p} a_{ik}\Delta f_{t-k} + \sum_{k=1}^{p} b_{ik}\Delta s_{t-k} + \left(\sum_{k=1}^{p} c_{ik}\Delta f_{t-k} + \sum_{k=1}^{p} d_{ik}\Delta s_{t-k} + \gamma_i z^*_{t-1}\right) \times W_i(z^*_{t-d}) + e_{it},$$

$$W_i(z^*_{t-d}) = 1 - \exp\{-\lambda_i \times (z^*_{t-d})^* \times g_i(z^*_{t-d})\},$$

$$g_i(z^*_{t-d}) = 0.5 + 1/[1 + \exp\{-\theta_i z^*_{t-d}\}],$$

	Sep 98		Dec 98		Mar 99	
	Futures	Cash	Futures	Cash	Futures	Cash
a_0	−0.002 (−1.14)	−0.001 (−0.84)	0.000 (0.01)	0.001 (0.87)	0.001 (0.81)	0.001 (−1.03)
a_1	−0.334 (−12.57)	0.337 (11.42)	−0.324 (−11.62)	0.343 (16.07)	−0.403 (−19.71)	0.194 (8.30)
a_2	−0.212 (−6.91)	0.189 (6.03)	−0.213 (−6.08)	0.162 (4.59)	−0.241 (−7.63)	0.124 (4.68)
a_3	−0.091 (−2.12)	0.114 (3.52)	−0.066 (−2.00)	0.131 (4.95)	−0.165 (−6.07)	0.103 (4.61)
a_4	−0.044 (−1.44)	0.055 (2.14)	−0.056 (−1.86)	0.057 (2.71)	−0.114 (−6.18)	0.071 (3.50)
a_5	−1.337 (−1.05)	−0.247 (−0.39)	0.317 (7.79)	0.440 (0.71)	−0.095 (−5.14)	0.041 (1.81)
a_6	−0.011 (−0.17)	0.114 (2.73)	−0.244 (−1.03)	0.007 (0.09)		
b_1	0.242 (7.75)	−0.305 (−9.34)	0.190 (5.47)	−0.391 (−14.51)	0.338 (14.41)	−0.190 (−6.66)
a_2	0.172 (5.51)	−0.182 (−5.47)	0.215 (6.09)	−0.120 (−3.71)	0.238 (6.92)	−0.100 (−3.23)
a_3	0.083 (1.77)	−0.095 (−2.80)	0.112 (3.43)	−0.096 (−3.56)	0.169 (5.70)	−0.081 (−3.82)
a_4	0.048 (1.66)	−0.039 (−1.36)	0.067 (1.93)	−0.044 (−1.84)	0.101 (3.14)	−0.095 (−3.73)
a_5	0.016 (1.01)	0.006 (0.39)	0.002 (0.01)	−0.004 (−0.32)	0.128 (4.81)	−0.038 (−1.43)
a_6	0.011 (0.72)	0.013 (0.89)	−0.003 (−0.25)	−0.004 (−0.32)		
c_1	0.535 (6.75)	0.070 (1.05)	1.308 (4.02)	0.343 (16.07)	0.148 (1.35)	0.070 (0.09)
c_2	−0.409 (−2.38)	−0.499 (−11.10)	0.021 (0.08)	0.162 (4.59)	0.127 (1.07)	0.448 (0.92)
c_3	−0.351 (−2.28)	−0.446 (−11.75)	−1.692 (−4.93)	0.131 (4.95)	0.134 (1.22)	1.055 (1.21)
c_4	0.121 (1.06)	0.087 (5.56)	−0.829 (−4.89)	0.057 (2.71)	0.106 (1.02)	−1.073 (−0.74)
c_5	0.706 (2.09)	−0.082 (−4.19)	0.495 (1.84)	0.440 (0.71)	0.145 (2.01)	−0.726 (−0.54)
c_6	0.761 (2.62)	0.284 (8.28)	−0.390 (−1.40)	0.007 (0.09)		
d_1	−0.557 (−3.33)	−0.463 (−4.31)	−0.983 (−2.05)	−0.391 (−14.51)	−0.129 (−1.04)	−1.521 (−1.10)
d_2	0.562 (3.63)	0.651 (7.58)	−0.336 (−2.37)	−0.120 (−3.71)	−0.106 (−1.07)	−1.717 (−2.14)
d_3	0.296 (0.57)	−0.323 (−7.44)	1.708 (3.65)	−0.096 (−3.56)	−0.089 (−0.85)	−0.885 (−1.11)
d_4	0.079 (0.71)	0.251 (8.02)	1.531 (5.23)	0.368 (1.07)	−0.090 (−0.77)	2.849 (1.27)

d_5	−1.230 (−3.17)	−0.244 (−7.79)	1.605 (3.49)	−1.660 (−3.52)	−0.206 (−2.29)	0.231 (0.18)
d_6	−0.300 (−6.45)	−0.182 (−3.94)	0.689 (2.21)	−1.126 (−2.26)		
γ	−0.510 (2.52)	1.193 (13.79)	−1.247 (−4.35)	1.657 (4.62)	−0.179 (−2.77)	1.734 (5.54)
λ	0.131 (4.08)	0.042 (3.01)	0.049 (4.87)	0.031 (7.31)	3.017 (2.61)	0.037 (2.06)
θ	−0.041 (−0.12)	−7.128 (−12.54)	−138.8 (−6.76)	−42.73 (−8.88)	−123.7 (−3.05)	−5.316 (−2.30)
EGARCH coefficients[c]						
ω	−0.054 (−4.11)	−0.065 (−2.56)	−0.069 (−2.60)	−0.078 (−3.66)	−0.237 (−68.21)	−0.203 (−123.7)
α	0.153 (7.16)	0.173 (6.97)	0.209 (7.14)	0.189 (8.64)	0.193 (13.08)	0.188 (−15.83)
β	0.986 (291.3)	0.985 (173.4)	0.980 (135.4)	0.980 (183.8)	0.942 (921.9)	0.953 (618.9)
δ	−0.390 (−5.72)	−0.321 (−7.19)	−0.160 (−3.53)	−0.080 (−1.89)	−0.131 (−0.65)	−0.169 (−2.91)
p-values of residuals diagnosis[d]						
Ljung–Box Q(24)-statistics						
u_t	0.659	0.214	0.693	0.864	0.961	0.643
u_t^2	0.608	0.314	0.199	0.096	0.632	0.025
Engle and Ng (1990) *Asymmetric tests*						
Sign test	0.751	0.340	0.051	0.284	0.661	0.218
Neg. size test	0.322	0.980	0.134	0.043	0.153	0.061
Pos. size test	0.803	0.879	0.582	0.547	0.230	0.879
Joint test	0.719	0.674	0.040	0.092	0.288	0.180

[a] We obtain the EC term z_t^* by running the following OLS:

$$f_t = c_0 + c_1 f_t^* + c_2 \times DEXP_t + z_t^*,$$

where $DEXP_t$ is the number of days before the expiration of the futures contract. We focus on three parameters of the STAR EC model, γ_i, λ_i, and θ_i. γ_i, which should be negative for the futures and positive for the cash, has a magnitude that is inversely related to the speed of information disseminated in the market. λ_i measures how quickly investors in market i respond to the mispricings. θ_i describes the asymmetric behavior of investors towards positive and negative mispricings.

[b] t-statistics are in parentheses.

[c] We allow the residuals to follow an exponential GARCH (EGARCH) process of Nelson (1991).

[d] The last panel reports the p-values of residual diagnosis, including the Ljung–Box statistics for 24th-order serial correlation in the level and squared standardized residuals as well as the asymmetric test statistics of Engle and Ng (1990). All the statistics indicate that the estimated STAR EC model reasonably fits the data.

5.2. Empirical results of the STAR error correction between DJIA futures and cash

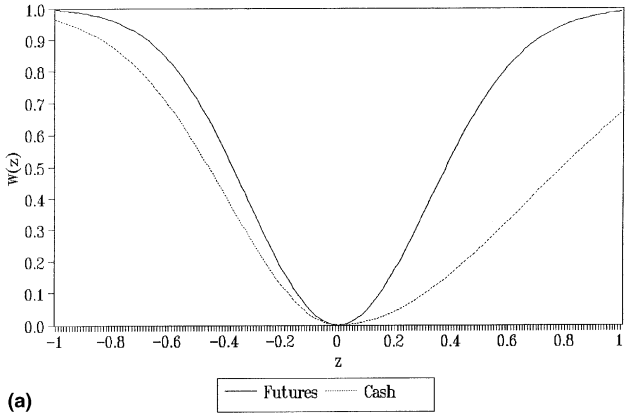
The lag lengths p, 6, 6, and 5, respectively, for the three contracts, are chosen by the Schwartz criterion (SIC). The number of parameters estimated for the STAR EC model are 32, 32, and 28, respectively. We report the results in Table 4 and focus on three parameters, γ_i, λ_i, and θ_i. γ_i, which should be negative for the futures and positive for the cash, has a magnitude that is inversely related to the speed of information disseminated in the market. λ_i measures how quickly investors in market i respond to the mispricings. θ_i describes the asymmetric behavior of investors towards positive and negative mispricings.

γ_is are significant with the correct signs in both markets of all contracts, suggesting a bidirectional error correction between the two markets. The three contracts in sequence have a significantly larger magnitude of γ_2 than γ_1, (−0.51, 1.19), (−1.25, 1.66), and (−0.18, 1.73), respectively. Thus, the DJIA futures market reflects information earlier than the stock market; consequently, mispricings occur. All three contracts show that λ_1 is greater than λ_2, and both are significantly positive: $(\lambda_1, \lambda_2) = (0.13, 0.04)$, (0.05, 0.03), and (3.02, 0.04). These results suggest that investors respond more quickly in the futures market than the stock market, and arbitrageurs establish the futures leg first.

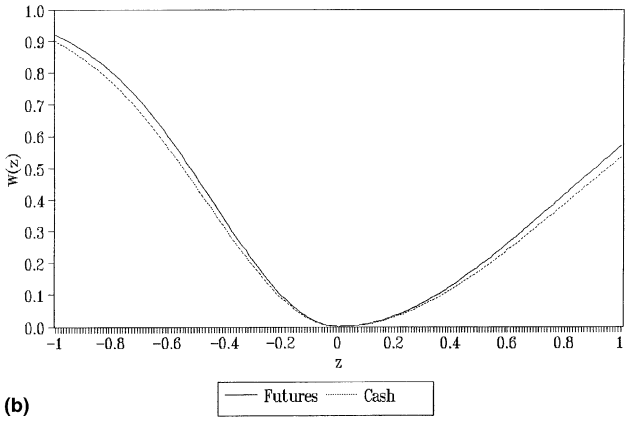
We plot the transition functions in Figs. 2(a)–(c). They show that the functions of the futures are steeper than those of the cash, although the difference is small for the Dec 98 contract. All the figures also demonstrate the markets' asymmetric responses to positive and negative mispricings. Since θ_is are negative in all periods and both markets, $(\theta_1, \theta_2) = (-0.04, -7.13)$, (−138.8, −42.7), and (−123.7, −5.32), investors respond more quickly to negative mispricings. That is, investors will trade more quickly when the futures price is less than the fair value. In the above, we have shown that the futures market reflects information faster. Thus, negative mispricing may convey a message to investors that the futures price is reflecting bad news, which makes the futures price drop. Accordingly, a negative θ_i suggests that more investors react to bad news than good news.

All the EGARCH parameters in Eq. (11) are significant in the three contracts. In particular, δ_is are negative, consistent with the notion that the markets are more volatile with bad news than good news. The last panel of Table 4 reports the p-values of residuals diagnosis, including the Ljung–Box statistics for 24th-order serial correlation in the level and squared standardized residuals as well as the asymmetry test statistics (sign bias, negative bias, positive size, and joint tests) developed by Engle and Ng (1990). Both the Ljung–Box and the asymmetry checks indicate that the estimated STAR EC model reasonably fits the data.

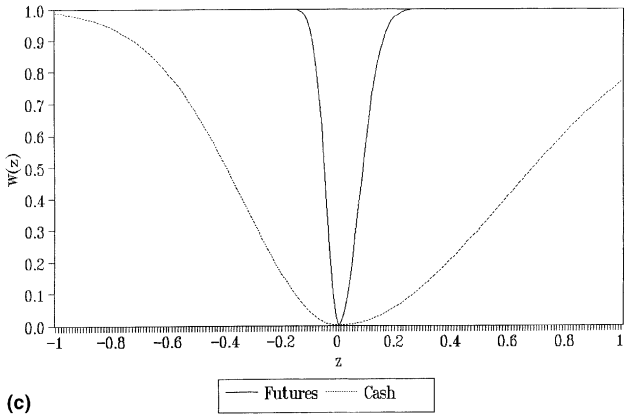
For comparison, we also estimate the linear EC model in (6) with EGARCH. The linear model imposes the condition that both markets adjust to



(a)



(b)



(c)

Fig. 2. Transition functions: (a) Sep 98; (b) Dec 98; (c) Mar 99.

any mispricing immediately, and assumes no legging in establishing the positions. The parameters of the EC terms, γ, are (−0.057, 0.077), (−0.073, 0.113), and (−0.106, 0.017). These results show that the futures market is moderately more efficient than the cash market during the first two contracts, but the cash market is much more efficient than the futures during the third contract. Therefore, results of the linear EC model cannot provide conclusive evidence as to which market reflects information more efficiently.

6. Conclusions

The basic index arbitrage model assumes homogeneous investors with a threshold mispricing between the futures and cash prices. The traditional model indicates that all investors exploit the arbitrage opportunities simultaneously when the mispricing exceeds the threshold. In contrast, we consider heterogeneity in investors' trading costs and objectives, capital constraints, and perceived risks of arbitrage. No single fair value is universally appropriate; different investors would exploit the opportunities at different mispricings. When we aggregate all the individual thresholds, the different thresholds will become a smooth transition function. We examine intraday index arbitrage of the DJIA with 5-minute intervals. All the 30 Dow stocks are actively traded on the NYSE. The DJIA is better than the S&P 500 index in studying the arbitrage mechanism because of less infrequent trading of component stocks and less tracking errors, execution risk, and transaction costs.

We model the mispricing as a smooth transition autoregressive (STAR) process. The STAR model allows for the fact that the larger the mispricing, the stronger the tendency to move back to equilibrium, and the transition is smooth in the aggregate. We show that the observed mean reversion in mispricing changes is induced by heterogeneous arbitrageurs, instead of the infrequent trading. In particular, according to the simulation experiment, the STAR model can explain the puzzle raised by Miller et al. (1994) that significant negative autocorrelation in basis changes still exists even after eliminating all observations outside the threshold of transaction cost bands.

We use a STAR error correction model to describe the nonlinear dynamics between the DJIA futures and index. This nonlinear model describes not only which market reflects information earlier, but also the speed of the market's reaction to mispricing. The linear error correction model is silent about the latter process and may provide biased results. The STAR EC model provides a rich description of index arbitrage, although the STAR model has not received substantial attention in finance. The model shows that information is reflected in the futures market first. When mispricing arises, more investors establish positions on the futures market before the stock market. We also find that, in both markets, investors respond more rapidly to negative mispricing than

positive mispricing, suggesting that investors react to bad news more than good news.

Acknowledgements

I thank the CBOT for providing me with the DJIA futures data and detailed information on the index arbitrage trading mechanism. I am grateful to the valuable comments and suggestions from Giorgio Szegö (the editor), two anonymous referees, John Broussard, Upinder Dhillon, Lisa Flynn, Srini Krishnamurthy, Dennis Lasser, Frank Samuel, Chun-chi Wu, Anjun Zhou, and seminar participants at Rutgers University at Camden and Syracuse University. This research is financially supported by the CBOT Educational Research Foundation. All errors are my own responsibility.

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